Machine Learning

DD2421, 7.5 credits

Atsuto Maki, André Pereira, Jörg Conradt

Autumn, 2025

1. About the course
   - Course Contents
   - Textbooks
   - Who are teaching?
2. Logistics
   - Lectures
   - Labs
   - Examinations
   - Miscellaneous
3. A very brief overview of Machine Learning
   - Applications
   - Types of Learning
   - Supervised and Unsupervised

The aim of the course is to provide:

- basic knowledge of the most important algorithms and theory that form the foundation of machine learning
- a practical knowledge of machine learning algorithms and methods

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Course contents:

- Lectures 1–11
- Lecture 12, mini lectures + exam Q&A
- LAB1 (labs 1–3, **NB. there is a deadline for each**)
- Written exam
- Programming challenge
  (part of the exam; → points contribute to your exam)

Intended outcomes – students will be able to:

- describe the most important algorithms and the theory that constitutes the basis for machine learning and artificial intelligence
- explain the principle for machine learning and how the algorithms and the methods can be used
- discuss advantages with and limitations of machine learning for different applications

in order to be able to identify and apply appropriate machine learning technique for classification, pattern recognition, regression and decision problems.

Recommended reading

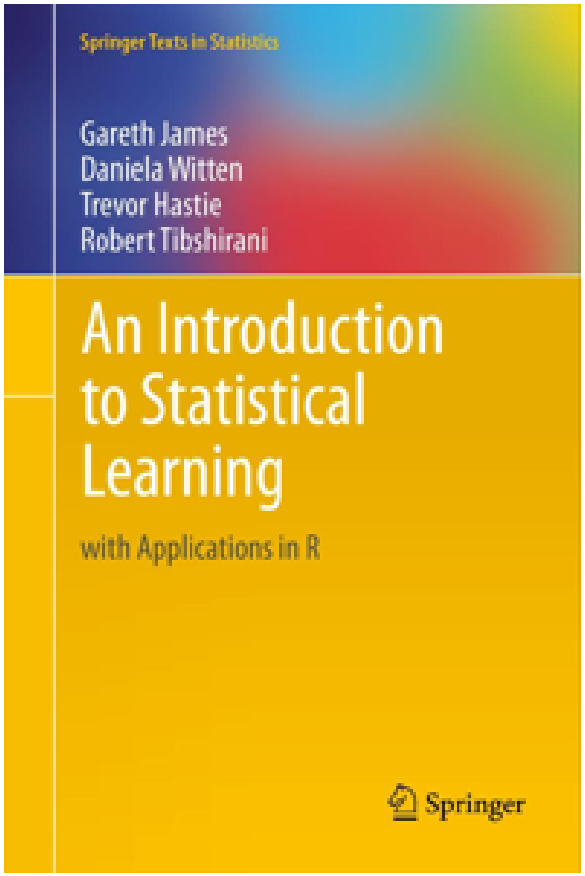
Gareth James, Daniela Witten,
Trevor Hastie and Robert Tibshirani

An Introduction to Statistical Learning

Springer, 2021

Available online:

https://faculty.marshall.usc.edu/gareth-james/ISL/



Recommended reading

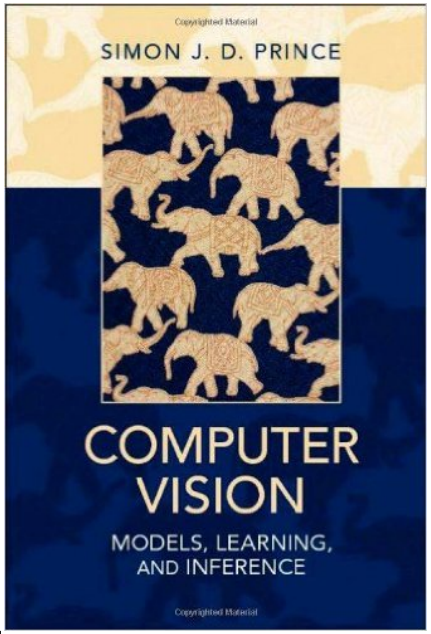
Simon Prince

Computer Vision: Models, Learning, and Inference

Cambridge University Press, 2012

Available online:

web4.cs.ucl.ac.uk/staff/s.prince/book/book.pdf



NB. Pointers to other reference materials under Lectures page.

Who are teaching?

- **Atsuto Maki**
  Div. Robotics, Perception, and Learning (RPL)
- **André Tiago Abelho Pereira**
  Div. Speech, Music, and Hearing (TMH)
- **Jörg Conradt**
  Div. Computational Science and Technology (CST)

- Course Assistant: **Alexander Kozlov**
  Div. Computational Science and Technology
- ca. 10 teaching assistants (the majority is PhD students)

André Tiago Abelho Pereira

Research Mission

From simple transactional, disembodied interactions



*To complex **social** embodied interactions*



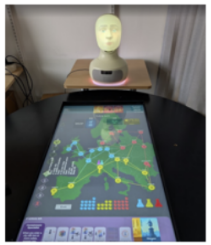
Research topics: Human-Robot Interaction, Multimodal Machine Learning

Example Projects:

fMRI as a modality to study conversational engagement with humans and robots



multimodal LLMs for collaborative embodied AI in games and education



robot telepresence for data collection and multimodal model creation

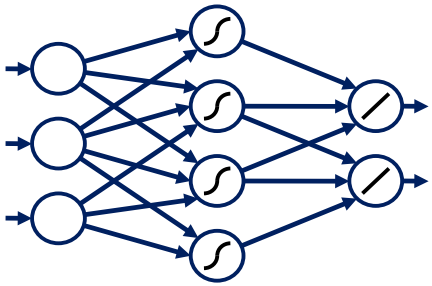
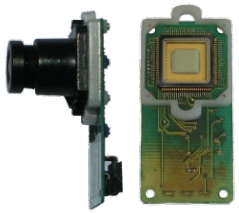
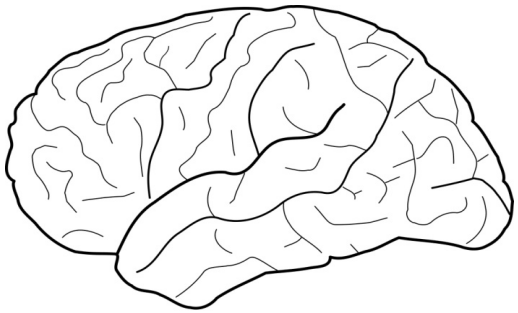
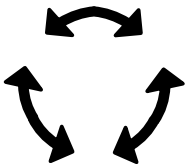


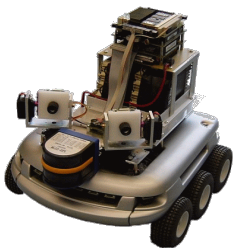
Teaching

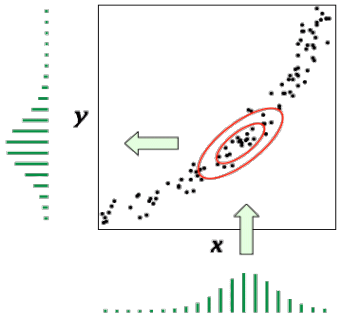
- Artificial Intelligence (Course responsible)
- Topics in Social Robotics (PhD Course)
- Machine Learning (this course)

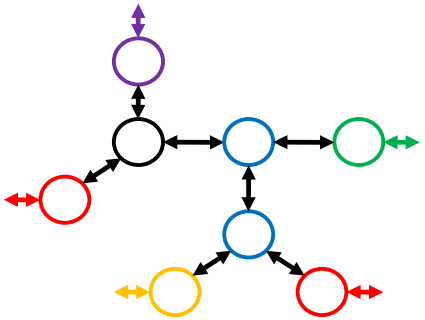
Jörg Conradt

Research Topic:
Neural Computation for Real-Time System Engineering







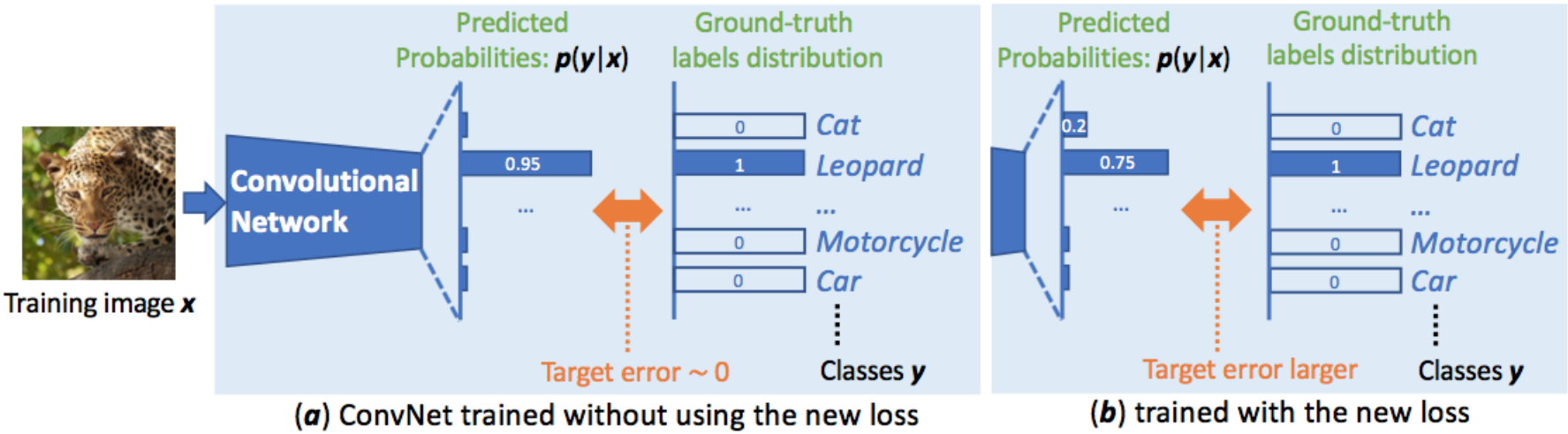


http://neurocomputing.systems

NCS

Atsuto Maki

Research topics: Computer Vision, Machine/Deep Learning



https://robotics.sciencemag.org/content/4/30/eaaw1329.full

https://www.kth.se/profile/atsuto

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Course Information on Canvas

https://canvas.kth.se/courses/56446

Course Information on KTH Social

https://www.kth.se/social/course/DD2421/
https://www.kth.se/social/course/DD2421/calendar/
https://www.kth.se/social/course/DD3431/

Course registration needed!

Any inquiries to be directed to Student service support EECS.

For administrative questions please consult this page:
www.kth.se/en/eecs/studentsupport

Lectures

1. Nearest Neighbour Classifier (Memory-based)
2. Decision Trees (Logical inference)
3. Challenges in Machine Learning
4. Regression
5. Probabilistic Methods
6. Learning as Inference
7. Learning with Latent Variables
8. Support Vector Machines
9. Artificial Neural Networks
10. Ensemble Methods
11. Dimensionality Reduction
12. Mini lectures (beyond the scope of DD2421), exam Q&A

LAB1 (3.5 credits)

1. Decision Trees
2. Support Vector Machines
3. Bayes Classifier & Boosting

- labs are carried out by students and examined by TAs
- use Canvas to book time slots before the deadlines

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Examination:

- It is your task to convince the examiner that you have done the assignment and understood the results.
- Strongly encouraged to work+report by pairs of 2 students (not 3!).
- 10 minutes, **be there on time**.
- No programming code to be shown
- Bring your ID (tell the TA if you are yet to be registered)

Examination (4 credits)

A written examination.
Date: Thu 16 October 14:00-18:00

Exam Registration in advance!

https://www.kth.se/form/exams
Registration opens Tuesday 16 September and closes Tuesday 30 September at 23.59. (please double-check!)

Set a reminder by yourself! (teachers cannot help if you missed it)

Chance for re-exam (Tue 16 Dec 14:00-18:00) - registration required.

Examination (4 credits) cont'ed: a programming challenge.

Build and train a classifier given a labeled dataset.
Use it to infer the labels of a given unlabeled evaluation dataset.

Submit the inferred labels - will be compared to the ground truth.
Provisional schedule (one week before the exam): 09-15 October

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The accuracy is proportional to the point you receive.

NB. You can figure out your point by yourself, pls do that!

Logistics will be made available on Canvas, under Assignments and/or exam Canvas:

FAQ

Q. Are course slides available?

A. Will be uploaded on Canvas under the "Lectures" in advance.

Q. Can you register me to the course / examination?

A. No, please consult Student service support EECS:
www.kth.se/en/eecs/studentsupport

Q. For taking the re-exam, do I need to do the RE-challenge?

A. Yes – your score of RE-challenge will be counted in.

PhD students

Please inform/specify your KTH-account by email with subject "DD3431 HT25" addressed to the examiner.
Furhter instructions for extra report will be provided.

If you are from another Swedish university, a form to get a KTH-account available at the reception of EECS.
See the instructions:
https://intra.kth.se/en/eecs/forskarutbildning/courses

Miscellaneous

* Message board available on “Discussion” on KTH Canvas (but bear with us – teachers cannot promise to respond :-)).

* Course committee: It will be a great pleasure to have students’ course committee (a.k.a. kursnämnd). Anyone volunteers, please?

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Sample Applications

- Image recognition / Computer vision
- Speech recognition and synthesis
- Natural language processing
- Autonomous robots
- Spam-filter for e-mail
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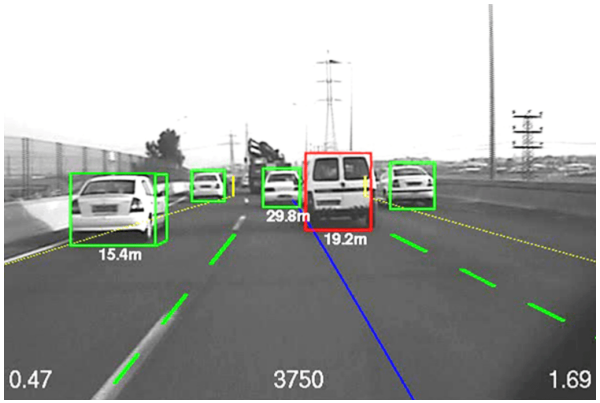
Where is machine learning useful?

A pattern exists

Data available for training

Hard/impossible to define rules mathematically

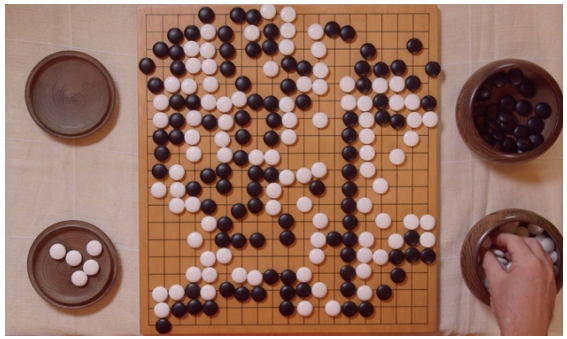
Driving assistants
(Google, Toyota, Volvo, . . .)



Personal assistants
(Apple Siri, Amazon Eco, . . .)



Board games
(DeepMind AlphaGo, . . .)



LLM, Translation
(OpenAI GPT, . . .)



Types of Learning

- **Supervised Learning** (covered)
- **Unsupervised Learning** (briefly covered)
- **Reinforcement Learning** (not covered)
- **Evolutionary Learning** (not covered)

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  - Function approximation
  - Well-defined problem. Battle-tested in industrial applications.
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  - Consider how a baby learns to walk for instance.
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- **Evolutionary Learning** (not covered)
  - General Purpose Optimization

Supervised vs Unsupervised learning paradigms

Conflates two different distinctions:

- **Supervised Learning**, a.k.a. predictive

- **Unsupervised Learning**, a.k.a. descriptive

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  - Learning mappings from A to B.
    (Neutral mathematics.)
  - Learning from human supervision: B was provided by a human teacher, as in "This is a dog".
    (Not scalable and biologically implausible.)

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- **Unsupervised Learning**, a.k.a. descriptive
  - Analyzing unstructured raw data. There is no B, only A.
  - Learning without human supervision.
    (Scalable and biologically plausible.)